

Power Of GPUs Today

Though CPUs once had four times as many transistors as GPUs, GPUs today have moved far ahead with their processing capabilities bringing supercomputing experience to the desktop. These can handle enormous data processing from standard-definition to high-definition, and from 2D to 3D

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Graphics processing units (GPUs) started getting into desktop computers for graphics enhancement a decade ago. Today, with their immense parallel computing capability, GPUs have made their way even to supercomputers. Nebulae—a supercomputer built on a cluster of 4640 blade servers—ranks as the second most powerful computer in the world in the June list of the fastest supercomputers by TOP500, a project that ranks most powerful computers in the world. It uses NVIDIA Tesla series general-purpose GPUs, exploiting the massively-parallel capabilities and providing unbelievable 3 petaflops/s peak capability—the highest ever on the TOP500.

Video and multimedia are today the centre stage for every user. Narendra Bhandari, director, Intel Software

and Services Group, says, “The average mainstream user is expecting a relatively rich visual experience in almost every platform. From a low-end core to a high-end system, the expectation of the user in terms of video quality or just user experience is always on a higher side.”

GPUs have enabled the best user experience from high-definition (HD) to three-dimension (3D), from the living room to personal computing.

“The key driving factors for GPUs are the three C’s—consumer, computing and communication,” says Nishant Goyal, head of sales, South Asia for NVIDIA. The evolution of consumers’ lifestyle with increasing usage of technology products such as set-top boxes, LCD TVs and digital cameras has forced the industry to provide easy and efficient solutions. Fusion of CPU and GPU has been successful in delivering supercomputing capability. These are

being used in applications like scientific image processing, spatial exploration, oil exploration, 3D reconstruction and even stock options price determination.

GPUs as a green supercomputer

In 2006, while delivering a keynote at *International Conference on Computer-Aided Design (ICCAD)*, AMD’s ex-chief technology officer Phil Hester stated that it was important to examine the capabilities of GPUs for the round two of killer micros. GPUs have moved from simple wire-frame rendering in the 1980s to 32-bit parallel-processing engines. Hester predicted that the combination of CPUs and GPU functions in heterogeneous cores would boost supercomputers’ performance, which is now a reality.

GPUs today are highly parallel general-purpose single-instruction multiple-data (SIMD) processor arrays. These not only boost the computing